Note: See pages 3-4 for an alogrithm and additional example.

Problem 1. Rewrite

$$\int_0^1 \int_0^{x^2} \int_0^y f(x,y,z)\,dz\,dy\,dx$$

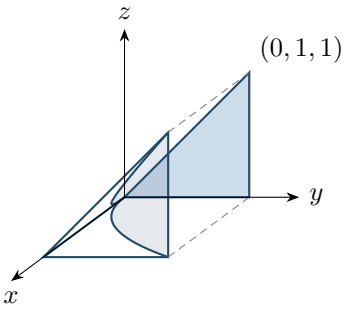
so that we integrate first with respect to x, then z, and then y.

Solution. The original bounds describe the region

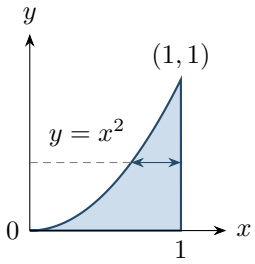
$$E = \{(x,y,z) : 0 \le x \le 1,\ 0 \le y \le x^2,\ 0 \le z \le y\}.$$

Since we integrate with respect to x first, we combine the restrictions on x. Notice that $y \le x^2$ and $x \ge 0$ imply $x \ge \sqrt{y}$. Since $\sqrt{y} \ge 0$, this is more restrictive than $x \ge 0$. Together with $x \le 1$, this yields that $\sqrt{y} \le x \le 1$.

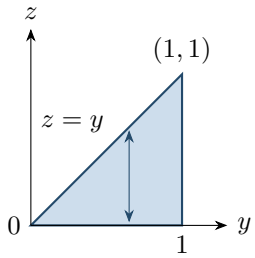
Solid and yz-shadow



xy-projection



yz-projection



We now project onto the yz-plane. For an x to satisfy $\sqrt{y} \le x \le 1$, we need $0 \le y \le 1$. Keeping the restriction $0 \le z \le y$, the projection is $P = \{(y,z) : 0 \le y \le 1,\ 0 \le z \le y\}$. Thus, the projection is the triangle bounded by $z = 0$, $z = y$, and $y = 1$. For each fixed y in this triangle, z ranges from 0 to y. Finally, the full range of y is $0 \le y \le 1$. Hence, the integral becomes

$$\int_0^1 \int_0^y \int_{\sqrt{y}}^1 f(x,y,z)\,dx\,dz\,dy.$$

Therefore, the correct answer is D.

Problem 2. Convert the integral

$$\int_{-3}^{3}\int_{0}^{\sqrt{9-x^2}}\int_{0}^{\sqrt{x^2+y^2}} 8(x^2+y^2)\,dz\,dy\,dx$$

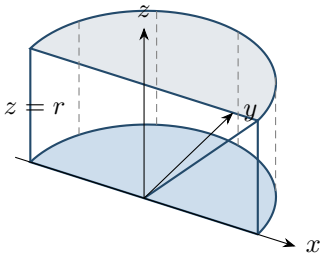
to cylindrical coordinates.

Solution. We first consider the projection onto the xy-plane. The bounds

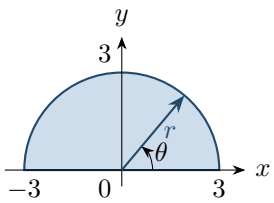
$$-3 \le x \le 3, \qquad 0 \le y \le \sqrt{9-x^2}$$

describe the upper half of the disk $x^2+y^2 \le 9$. Thus, in polar coordinates, this projection is given by $0 \le r \le 3, 0 \le \theta \le \pi$. The solid lies above $z=0$ and below the cone $z = \sqrt{x^2+y^2} = r$. Hence, the remaining bounds are $0 \le z \le r$.

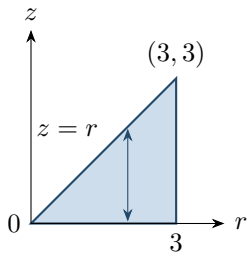
Solid Outline and xy-shadow



xy-projection



rz-cross-section



The shaded semicircle is the shadow of the solid on the xy-plane. For each fixed angle θ, the rz-cross-section is the triangle $0 \le z \le r \le 3$. Finally, $x^2+y^2=r^2$ and $dV = r\,dz\,dr\,d\theta$. Therefore,

$$8(x^2+y^2)\,dV = 8r^2 \cdot r\,dz\,dr\,d\theta = 8r^3\,dz\,dr\,d\theta,$$

and the integral becomes $\int_0^{\pi}\int_0^3\int_0^r 8r^3\,dz\,dr\,d\theta$. Therefore, the correct answer is D.

Choosing Bounds in a Given Order

Suppose the requested order is $dx\,dz\,dy$. We find the x-bounds first, then the z-bounds, and finally the y-bounds.

1. **Write the region as inequalities.** If an integral is given, write each pair of bounds as an inequality. If the solid is described by surfaces, translate the description into inequalities: for example, above $z = 0$ means $z \geq 0$, and below $z = y$ means $z \leq y$. Keep all these inequalities together. Every point in the solid must satisfy all of them.

2. **Combine the restrictions on x.** Treat y and z as fixed numbers for the moment. Rearrange the inequalities involving x and combine them to find its allowed range. Use the largest lower bound and the smallest upper bound. These are the most restrictive bounds: they make all the restrictions on x hold at once. Their equations describe the surfaces bounding the solid in the x-direction.

3. **Find the shadow on the yz-plane.** Now ask: for which pairs (y, z) is there an allowed value of x?

   If the x-bounds are $L(y, z) \leq x \leq U(y, z)$, then we require $L(y, z) \leq U(y, z)$ in addition to any conditions needed for these expressions to be defined. Keep all restrictions involving only y, z (These describe the projection, or shadow, on the yz-plane.)

4. **Combine the restrictions on z in this shadow.** Treat y as fixed. Find the smallest and largest allowed values of z using the inequalities for the shadow. These bounds can depend on y, but cannot contain x or z.

5. **Find the full range of y.** Determine which y-values allow at least one z in the shadow. These give the constant outer bounds. Write the integral in the requested order:

$$\int_a^b \int_{g_1(y)}^{g_2(y)} \int_{h_1(y,z)}^{h_2(y,z)} f(x, y, z)\,dx\,dz\,dy.$$

Note: If an allowed range consists of separate intervals, or the formula for a boundary changes across the region, split into pieces and use a sum of integrals. Do not fill in gaps between separate intervals. Additionally, your bounds should describe exactly the original solid, and for another integration order, relabel the variables and follow the same steps.

Example

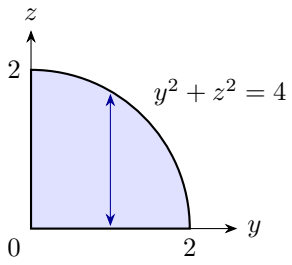
Set up the volume integral for the solid in the first octant between $x = y^2 + z^2$ and $x = 4$, in the order $dx\,dz\,dy$.

1. The first octant gives $x, y, z \geq 0$. Between the paraboloid and the plane, x is at least $y^2 + z^2$ and at most 4. Thus, $x \geq 0, y \geq 0, z \geq 0, x \geq y^2 + z^2, x \leq 4$.

2. For fixed y, z, we have two lower bounds, 0 and $y^2 + z^2$, and one upper bound, 4. Since $y^2 + z^2 \geq 0$, the stronger lower bound is $y^2 + z^2$. Therefore, $y^2 + z^2 \leq x \leq 4$.

For example, fixing $y = 1$ and $z = 1$ gives $2 \leq x \leq 4$. Choosing $x = 0$ would satisfy $x \geq 0$, but would violate $x \geq y^2 + z^2$.

3. An x can lie between $y^2 + z^2$ and 4 exactly when $y^2 + z^2 \leq 4$. Keeping $y, z \geq 0$, the shadow is:



4. For fixed y, we have $z \geq 0$ and $z^2 \leq 4 - y^2$. Hence, $0 \leq z \leq \sqrt{4 - y^2}$.

5. We need $4 - y^2 \geq 0$ and $y \geq 0$, so $0 \leq y \leq 2$. Therefore, the volume is $V = \int_0^2 \int_0^{\sqrt{4-y^2}} \int_{y^2+z^2}^4 1\,dx\,dz\,dy$.

Practice Problems

1. Set up the volume integral for the solid in the first octant bounded by the coordinate planes and the plane

$$2x + y + z = 6,$$

using the order $dx\,dz\,dy$.

2. Set up the volume integral for the solid

$$E = \{(x, y, z) : x^2 + z^2 \leq 4,\ 0 \leq y \leq z\},$$

using the order $dx\,dz\,dy$.

3. Rewrite the integral

$$\int_0^1 \int_0^1 \int_0^{x+y} f(x, y, z)\,dz\,dy\,dx$$

in the order $dx\,dz\,dy$.